



ME 300: Engineering Thermodynamics

Example: Work

Example: A closed piston-cylinder system filled with 1 kg of air starts at a volume of 1 m^3 and isobarically expands to 2 m^3 . If the initial temperature is 300 K, find the following and draw the process on a P-V diagram:

- P_2
- T_2
- ${}_1W_2$

Assume: ideal gas
: quasi-steady

Now what if I do that same process isothermally?

Isobaric process

$$V_1 = 1 \text{ m}^3 \quad V_2 = 2 \text{ m}^3$$

$$T_1 = 300 \text{ K} \quad P_2 = P_1$$

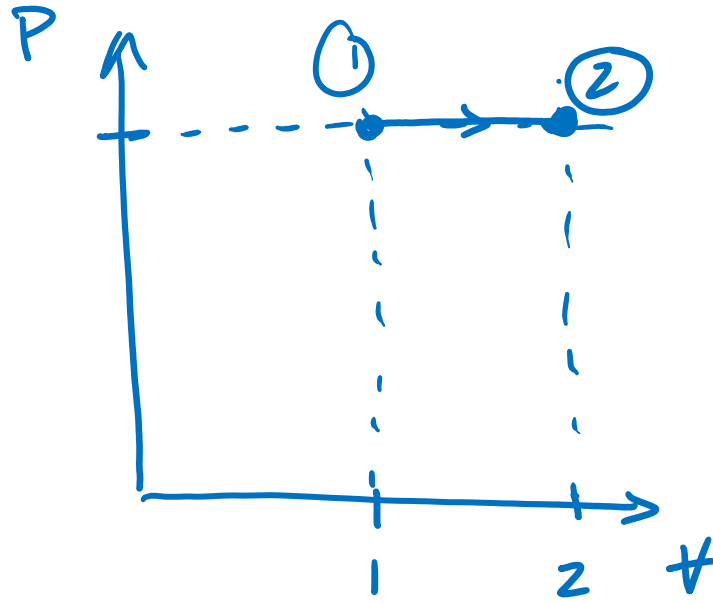
$$M = 1 \text{ kg}$$

$$P_1 V_1 = M R T_1$$

$$P_1 = \frac{M R T}{V}$$

$$= \frac{(1)(287)(300)}{1}$$

$$P_1 = 86100 \text{ Pa}$$



on? by?
+

$${}_1\bar{W}_2 = \int_{V_1}^{V_2} P dV = P \int_{V_1}^{V_2} dV = P(V_2 - V_1) = 86100(2 - 1)$$

$$\boxed{{}_1\bar{W}_2 = 86100 \text{ J}}$$

Isothermal process

$$V_1 = 1 \text{ m}^3 \quad V_2 = 2 \text{ m}^3$$

$$T_1 = 300 \text{ K} \quad T_2 = T_1 = 300 \text{ K}$$

$$P_1 = 86100 \text{ Pa}$$

$$M = 1 \text{ kg}$$

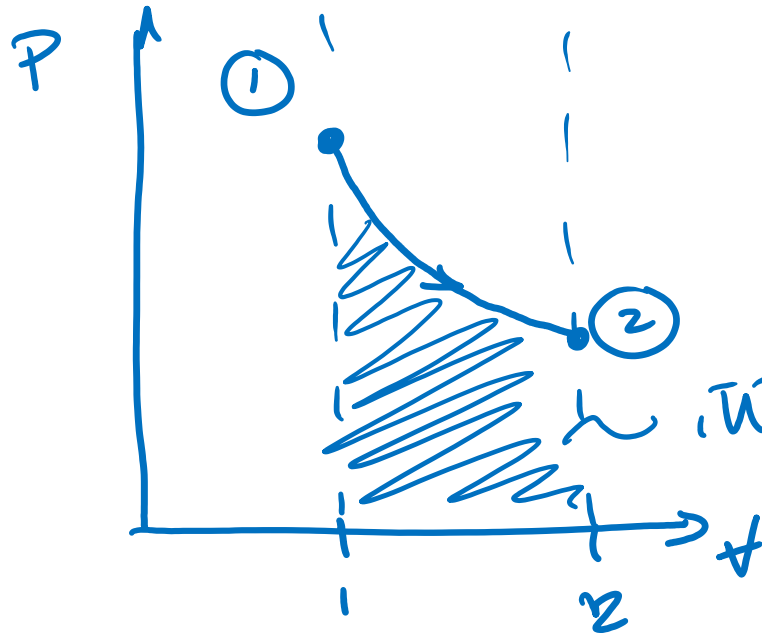
$${}_1\bar{W}_2 = \int_{V_1}^{V_2} P dV$$

$$P = \frac{MRT}{V}$$

$$= \int_{V_1}^{V_2} \frac{MRT}{V} dV = MRT \int_{V_1}^{V_2} \frac{dV}{V} = MRT \ln\left(\frac{V_2}{V_1}\right)$$

$$= (1)(287)(300) \ln\left(\frac{2}{1}\right)$$

$$\boxed{{}_1\bar{W}_2 = 59679.97 \text{ J}}$$



$${}_1\bar{W}_2 = \int_{V_1}^{V_2} P dV$$

Application to a cycle

1-2: isothermal compression $\Delta T = 0$

2-3: isochoric heat addition $\Delta V = 0$

3-4: isothermal expansion $\Delta T = 0$

4-1: isochoric heat rejection $\Delta V = 0$

$$1-2: {}_1\bar{W}_2 = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{MRT}{V} dV < 0$$

$$2-3: {}_2\bar{W}_3 = \int_{V_2}^{V_3} P dV = 0$$

$$3-4: {}_3\bar{W}_4 > 0$$

$$4-1: {}_4\bar{W}_1 = 0$$

$$\bar{W}_{\text{net}} = {}_1\bar{W}_2 + {}_3\bar{W}_4 > 0$$

☆ Power cycle $\rightarrow$ clockwise

